Supplement: constructions, reference data, and experiments

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This supplement accompanies *Controlling two-qubit involutions: CNOT costs and symbolic invariants*. It contains the reference resource, all measured compiler comparisons and negative results, native-column gap audits, and implementation details. Its empirical comparisons are separate from the interaction-count theorems in the main paper.

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1 Symbolic constructions and reference resources

The implementation accepts exact matrices and named symbolic families. It enumerates all distinct nontrivial cuts, factors rank-one realignments, and peels candidate interactions in a bounded residual search. Exact exponential-polynomial arithmetic supplies rank witnesses rather than numerically thresholded SVDs. Native controlled, exchange, B, and SQiSW charts supply complementary candidates. Every accepted candidate is checked by an exact identity or a compositional certificate of proved replacements. A failed search remains unresolved, and a search deadline is not an impossibility argument.

The B and SQiSW charts include folded-angle and degenerate branches. Long expressions are stored as a scalar directed acyclic program with scoped aliases. The canonical three-SQiSW projective chart has 585 scalar bindings and 11 physical gates. This is a machine-inspectable exact expression program, not a short human-readable formula. Numerical evaluation compiles the shared dependencies once and evaluates only the branches required for a binding; the exact family verifier remains the acceptance criterion. The numerical binding experiment in this supplement measures this implementation without claiming that symbolic synthesis is always faster than numerical synthesis.

The grid contains 327 catalog entries crossed with 32 supplied alphabets. There are 217 named one-qubit entries, including 192 matrices obtained from 24 Clifford representatives and eight phases; the remaining named widths are 72 two-qubit, 14 three-qubit, three four-qubit, and one five-qubit entry. Twenty register fixtures use specified widths. Their physical widths produce 218, 79, 26, 3, and 1 entries at widths one through five. Aliases and phase multiples are included explicitly and are not independent families. In literal mode, 222 entries admit zero-entangler constructions in every preset where they are representable; zero-cost nonlocal synthesis is not inferred from the size of the headline grid.

The Clifford reference includes finite-alphabet shortest-path proofs for $\{H, S, T, T^\dagger\}$: a radius-seven search reaches 628 matrices, and meet-in-the-middle splitting proves word minima for all 192 targets, with maximum length fourteen. Continuous-alphabet circuits have no corresponding unrestricted local-optimality claim. The result is a useful explicit phase reference alongside established Clifford+T synthesis theory [15, 16, 17, 18]. The actual gate and Clifford PDFs, their editable LaTeX, alphabetical index, and exact expression glossaries are included in the full review package.

2 Evaluation

2.1 Coverage by width and proof scope

The following tables classify every effective cell, including a literal circuit reused as its projective construction. "Fixed min." and "Family min." are a disjoint union of matching explicit lower-bound payloads. The latter certifies a necessary uniform family budget, not the minimum at every parameter specialization. "Unproved" means that a positive-entangler construction lacks a matched certificate in this union. Legacy Boolean optimality flags alone are not counted as proof payloads. Zero-entangler constructions and obstructions appear separately.

Literal equality.

Qubits	Cells	$E = 0$	$E > 0$	Fixed min.	Family min.	Unproved	Obstructed
1	6976	5210	0	0	0	0	1766
2	2528	117	1754	707	710	337	657
3	832	80	468	43	0	425	284
4	96	0	59	0	0	59	37
5	32	0	14	0	0	14	18
Total	10464	5407	2295	750	710	835	2762

Projective equality.

Qubits	Cells	$E = 0$	$E > 0$	Fixed min.	Family min.	Unproved	Obstructed
1	6976	6915	0	0	0	0	61
2	2528	128	2247	954	955	338	153
3	832	96	692	48	0	644	44
4	96	0	89	0	0	89	7
5	32	0	29	0	0	29	3
Total	10464	7139	3057	1002	955	1100	268

The row ledger lists gate, alphabet, phase policy, width, interaction count, proof scope, and every matched proof source. This resolves the earlier inconsistency between per-result flags and separate audit totals. In the pre-revision archive the raw positive-entangler flags were 694 literal and 423 explicitly stored projective results. Its 1,916 explicitly stored positive-entangler projective results omitted 1,141 positive-entangler literal circuits reused projectively. The denominator for the complete projective grid is therefore 3,057, not 1,916. Neither the flags nor an uncorrected sum of overlapping audits is our reported optimality total.

The obstruction breakdown is as follows.

Obstruction	Literal	Projective
determinant_and_clifford_t_field	0	16
determinant_group	2601	0
discrete_alphabet_nonconstant_family	142	210
not_a_clifford	13	38
outside_clifford_t_number_field	6	4

These are standard necessary conditions, used to describe feasibility. In particular, 2,601 literal obstructions are determinant-group tests, and all 268 projective obstructions occur in the three discrete Clifford or Clifford+T presets. A nonconstant continuous family cannot be covered by a countable discrete alphabet, although special parameter values can be representable. We do not present these simple feasibility checks as new lower-bound theorems.

2.2 Independent compiler comparisons

Eight fixed targets are constructed independently from library gates. Qiskit 2.6.2, pytket 2.18.5, and BQSKit 1.2.1 receive these source words; for targets on at most three qubits they also receive the target matrix. The table selects the smallest verified (E, L) pair over those recorded routes. The source is never our optimized output. Qiskit uses optimization level three and seed 219; tket uses full peephole optimization with implicit swaps disabled; BQSKit uses level three, maximum synthesis size three, and a 120-second process budget. All inputs are arbitrary states, ancilla-free, with fixed output wires and all-to-all connectivity. Full-matrix projective replay requires maximum entry error below 10^{-8}.

Quartz and the original Java-17 QUESO implementation receive the same independent sources in the NAM alphabet $\{H, R_z, X, \mathrm{CX}\}$. Each has a 10-second search budget and optimizes total native gate count. Quartz uses its supplied NAM equivalence set; QUESO rules are generated with three wires and length six. Rule files, repository revisions, the Quartz harness patch, and output QASM are archived. Their displayed local counts use one-qubit fusion into U3, preserving the entangling skeleton; these settings do not constitute an exhaustive search for each tool's best possible result.

Gate	Qiskit	tket	BQSKit	Quartz	QUESO	Exact atlas
SYC	3/8	3/7	3/4	4/5	4/5	3/3
iSWAP	2/3	2/4	2/4	2/3	2/3	2/3
SSWAP	3/8	3/7	3/5	–	3/8	3/7
fFredkin	5/13	10/15	4/6	9/10	13/10	4/6
CCX	6/8	6/8	6/8	6/8	6/8	6/8
CSWAP	7/10	7/8	7/9	8/8	8/8	7/10
C3X	14/19	14/16	14/16	14/16	14/16	14/17
C4X	36/41	36/41	–	30/33	36/36	36/41

Entries show E/L; "–" means no output passed the declared replay threshold on the recorded route. BQSKit has one runtime error, one 120-second exhaustion, and an unverified 30-CX C4X output. Quartz's SSWAP output has error about 6.9×10^{-7} and is excluded. Every failure and its settings remain in the raw record.

BQSKit independently matches our fFredkin count of four CNOTs and six locals. A separate enumeration of all 27 three-edge and 81 four-edge layouts, with three BQSKit starts per layout, also finds a four-CNOT fit in about six seconds overall. Failed numerical three-CNOT fits are not the lower-bound proof. The eleven older SDK entangler wins were all fFredkin in different presets; they were not eleven distinct gate discoveries. The current data also contain genuine losses: Quartz reaches 30 CNOTs for C4X while this atlas entry has 36; tket and BQSKit use fewer C3X locals; BQSKit uses fewer SSWAP locals. Consequently we claim neither compiler dominance nor that all atlas cells attain the best known upper bound.

The older ancilla-free SDK sources use 14 CX for C3X and 36 for C4X [19, 20]; 36 is a historical construction, not a best-known claim. We also checked the no-helper-qubit constructions of Huang–Palsberg [25] and Vale et al. [26], through their Qiskit 2.6.2 implementations. For C4X, full literal replay verifies 52/47 CX (levels zero/three) for HP24, and 36/36 for V24 and the existing SP22 route. The archived fixed C4X source is also 36/36. These particular routes do not beat Quartz's verified 30-CX result; asymptotic general-purpose bounds do not establish the smallest four-control count. The bounded literature/implementation audit is not an exhaustive best-published census. Ordinary Fredkin has a known seven-CX word [21]. Toffoli's six-CX minimum is proved [14]; the five-unrestricted-gate Fredkin result [13] uses a different primitive. UniversalQCompiler is the only listed analytical package without a run here, because its licensed Wolfram kernel was unavailable.

2.3 Component ablations

Six opaque two-qubit families are tested in B/U3 and SQiSW/U3: X/Y-controlled rotations, unequal controlled SU(2) branches, a phase-gauged exchange block, a gauged affine Cartan family, and unequal commuting exchanges. An earlier chart-disabled experiment verifies 3/12, compared with 11/12 for the full method. This establishes that native charts account for much of this workload's success.

We now separately disable OSR factor construction and residual interaction peeling, retaining the native charts, Clifford and commuting-Pauli constructors, and posthoc identity/lower-bound checks. Both methods receive opaque matrices and a 20-second outer deadline, with a 0.2-second residual search budget where enabled. The additional three- and four-qubit probes are explicitly defined independent-product circuits, not catalog names.

Qubits	Method	Cases	Verified	Unresolved	Deadline	Errors
2	Full	12	11	0	1	0
2	OSR construction off	12	11	0	1	0
3	Full	1	0	1	0	0
3	OSR construction off	1	0	1	0	0
4	Full	1	0	1	0	0
4	OSR construction off	1	0	1	0	0

Figure 1 displays these results. OSR construction does not increase the success count on the two-qubit chart workload, and neither method resolves the wider probes within this configuration. This experiment does not support attributing the chart successes to OSR search. OSR's role in the mathematical lower bounds is established analytically, independently of the ablation. The main paper accordingly attributes its mathematical results to the analytical arguments.

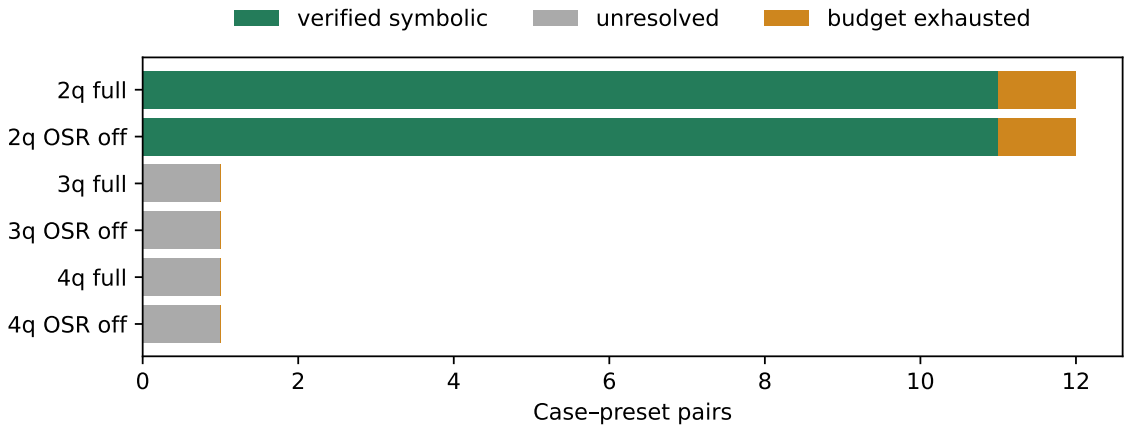


Figure 1: The full constructor and the constructor with OSR factorization and residual search disabled. The two-qubit workload is dominated by native charts. Wider unresolved probes are retained.

2.4 Parameter reuse and binding cost

The four-qubit example composes six fSim blocks on $(0,1),(2,3),(1,2)$ repeated twice, with twelve independent parameters. A gatewise source $R_{xx}(\theta)R_{yy}(\theta)CP(-\phi)$ is compared with a three-CX analytical source, using the same Qiskit 2.2.3 optimization. Three-CX fSim synthesis is established analytical knowledge, not a new resource bound.

Route	E	L	Depth	Compile (ms)	1,000 binds (ms)
Gatewise analytical source	36	66	55	50.76	114.28
Three-CX symbolic source	18	42	31	7.13	52.14
Numerical compile per binding	18	40	25	7.68	–

The numerical row is the mean compilation time over ten independently bound instances; all ten give 18/40/25. It beats the symbolic circuit in locals and depth. The weak gatewise 36/66/55 row is retained for context and is not evidence of improvement over the known three-CX construction. Using that known analytical source as a compile-once control gives the same interaction budget as the symbolic route. Qiskit already supports parameter reuse [8]. Cold symbolic chart synthesis took 5.63 seconds in this run; the displayed SDK time excludes that cost. Ten complete 16-by-16

numerical replays check the composed family, whose constituent identities are separately proved exactly.

For the larger 585-binding scalar DAG, compilation took about 1.36 seconds and 1,000 warm evaluations about 0.97 seconds. Full two-qubit matrix replay passed for 64 parameter assignments, including the identity, an Ising corner, the SWAP-class corner, and 60 seeded points spanning multiple angle folds. These are numerical regression tests of an exactly certified chart. They show that a large symbolic program can be reused without repeated symbolic substitution; they do not establish a speed advantage over bind-time KAK on arbitrary workloads. Symbolic outputs are valuable when a free-parameter identity, phase contract, or reusable expression is needed; specialization and numerical optimization remain useful alternatives.

2.5 A benchmark-suite peephole check

We apply a deliberately simple atlas peephole pass to every existing IBMEagle and QASMBenchmarks QASM source with two to eight declared qubits and fewer than 500 semicolons, yielding eight circuits. Numerical fingerprints propose contiguous windows of at most eight operations on two wires; exact identities accept matches. The control is gatewise atlas lowering into B/U3 or SQiSW/U3, with the same lowering on unmatched gates. Every substitution is therefore compositionally exact. The materialized native words also pass 32 seeded statevector replays. Source hashes, selection rules, counts, and attempted-match timing are archived.

Circuit	Basis	Qubits	E before	L before	E after	L after	Rules
QFT8	B	8	112	316	112	316	0
QFT8	SQiSW	8	112	316	112	316	0
barenco_tof_3	B	5	48	146	48	146	0
barenco_tof_3	SQiSW	5	48	146	48	146	0
barenco_tof_4	B	7	96	290	96	290	0
barenco_tof_4	SQiSW	7	96	290	96	290	0
mod5_4	B	5	56	163	56	163	0
mod5_4	SQiSW	5	56	163	56	163	0
tof_3	B	5	36	111	36	111	0
tof_3	SQiSW	5	36	111	36	111	0
tof_4	B	7	60	185	60	185	0
tof_4	SQiSW	7	60	185	60	185	0
adder_n4	B	4	20	53	20	53	0
adder_n4	SQiSW	4	20	53	20	53	0
tfim	B	4	120	310	120	310	0
tfim	SQiSW	4	120	310	120	310	0

The pass finds no improving windows on this suite. A separate positive control replaces the standard three-CX SWAP word by its archived native construction and verifies the expected interaction reduction. This checks the exact SWAP identity and native resource accounting, but does not turn the negative suite result into an application speedup. Wider windows, commutation search, and richer parameterized matching are future optimizer work. The present evidence supports the reference library and theory, not an end-to-end peephole performance claim.

3 Native-column corrections and remaining gaps

The SYC primitive is the catalog matrix $\mathrm{fSim}(\pi/2, \pi/6)$, with off-diagonal entries $-i$ and $|11\rangle$ phase $e^{-i\pi/6}$. The previous CNOT fallback used six SYC pulses. The exact chart below reduces controlled

phases and controlled-$SU(2)$ blocks to two. A separate literal DCX word uses one unrestricted-angle TK2, consistent with the Cartan capacity of that preset. Local gates are lowered into each supplied alphabet, with no physical GP gate. The unrestricted-angle TK2 capacity is one macro for every two-qubit Cartan class [27]. Other TK2 cells still exceeding one are annotated against that analytic cap. For literal equality it applies to the determinant-one targets representable by this alphabet; projective capacity does not remove a determinant obstruction. This capacity statement does not supply missing exact local frames in the implementation.

Preset	Modes	Count leads	Fixed gaps	Sample gaps	Analytic gaps
google_syc	150	150	24	35	0
sqrt_swap_u	150	48	14	2	0
tket_tk2	158	120	0	0	23

Each mode above is a literal or projective cell. A shorter numerical witness is annotated directly on the atlas panel and recorded in `native_column_gaps.csv`. For a fixed target, it is a numerical construction at that target; for a free family, it applies only to the explicit parameter substitution in the record. It is neither an exact symbolic construction nor a minimum proof. Counts already matched by an exact construction have no remaining gap. Searches that find no witness remain inconclusive.

The numerical audit uses the actual catalog SYC and square-root-SWAP matrices, including the latter's exchange-phase convention. It groups proposals by numerical Cartan coordinates, fits two then three constant native pulses with arbitrary local frames (five seeded starts per count), transports the frames to each original target, lowers them to the actual local alphabet, and absorbs scalar phase through allowed local gates. Every accepted witness passes full 4×4 literal replay with maximum entry error below 10^{-8}. All operations, parameter substitutions, replay errors, unsuccessful fits, seed, and versions are stored separately from the exact archive. The audit finds witnesses for all 75 nonlocal two-qubit SYC entries and 24 of the 75 square-root-SWAP entries. A free-family witness is only one sample. Failure on the other square-root-SWAP entries supplies no lower bound. Cirq already provides efficient native SYC decompositions for several of these classes [24]; improved counts are not claimed as discoveries.

3.1 Exact two-SYC controlled-phase chart

Put $S = \mathrm{SYC}$, $w = e^{-i\pi/6}$, $C = \cos(\pi/12)$ and $0 \le t \le \pi$. Define

$$c = \frac{\sin(t/4)}{C}, \quad s = \sqrt{1-c^2}, \quad \eta = 2\arccos c, \quad \varphi = 7\pi/12, \quad \beta = \mathrm{atan2}(4sC, c).$$

For this domain $s > 0$. With the second wire as control, $SR_x(\eta)_1 S = \mathrm{diag}(M_0, M_1)$, where

$$M_0 = \begin{pmatrix} c & -s \\ -s & -c \end{pmatrix}, \qquad M_1 = \begin{pmatrix} -c & -sw \\ -sw & cw^2 \end{pmatrix}.$$

Direct multiplication shows $M_0^2 = I$ and

$$M_0 M_1 = wVR_z(t)V^\dagger, \qquad V = R_z(\varphi)R_y(\beta).$$

Thus the following word, in time order, is literally $CP(t)$:

$$R_y(\beta)_0,\ R_z(\varphi)_0,\ S,\ R_x(\eta)_1,\ S,\ Z_0,$$
$$R_y(-\eta)_0,\ R_z(-\varphi)_0,\ R_y(-\beta)_0,\ P(t/2+\pi/6)_1.$$

An exact Pauli-coordinate proof verifies the relative-block identity using $c = \sin(t/4)/C$, $s^2+c^2 = 1$, and $c^2 + 16s^2C^2 = 16C^2\cos^2(t/4)$. The nonnegative axis coordinates fix the `atan2` branch. The certificate checker regenerates every operation and rejects a changed angle, phase, or score.

For a free real angle θ, let $q = \operatorname{atan2}(\sin\theta, \cos\theta)$, $t = |q|$, and $b = 1$ for $q < 0$, otherwise zero. Conjugate $CP(t)$ by X_1^b and append $P_0(-tb)$, using $X_1 CP(t) X_1 = P_0(t)CP(-t)$. This gives all signs and periods without a scalar operation. The exchange echo is consistent with Cirq's known analytical construction [24]; the supplied chart makes its local frames exact and explicit.

4 Implementation objective and reproducibility

5 Secondary objective and interpretation

The implementation orders candidate resources as (E, S, L), where E is native entanglers, L is local gates, and the optional description score is

$$S(C) = \sum_g \left[2^{k_g} + \frac{1}{5} \sum_{p \in \text{params}(g)} \left(\log_2 \max\{1, |\operatorname{den}(\text{simplify } p)|\} + \frac{1}{100} \right) \right]. \tag{1}$$

This heuristic prefers shorter parameter descriptions and named gates over unnecessary Euler parameters. It depends on the simplifier and has no calibrated pulse-error interpretation. The theorems optimize E alone. Our compiler table compares (E, L) explicitly, independently of this score; more locals can occasionally score better under the archive's objective. Unknown equal-E scores remain incomparable, not zero. The nonnegative score is a configurable description heuristic, not an optimality theorem.

6 Preset definitions

Preset	Entangler	Local alphabet
cx_u	CX	U3
cx_rz_ry	CX	Rz, Ry
cx_h_phase	CX	H, P
ibm_cx	CX	Rz, Sx, X
ibm_ecr	ECR	Rz, Sx, X
ibm_cz	CZ	Rz, Sx, X
cz_u	CZ	U3
cz_rz_rx	CZ	Rz, Rx
iswap_u	iSWAP	U3
sqrt_iswap_u	SiSWAP	U3
sqrt_swap_u	SSWAP	U3
xx90_u	XX90	U3
xx_rz_ry	Rxx	Rz, Ry
zz_u	Rzz	U3
zx_u	Rzx	U3
google_cz	CZ	PhasedXZ
google_sqrt_iswap	SiSWAP	PhasedXZ
google_syc	SYC	PhasedXZ
google_fsim	fSim	PhasedXZ
berkeley_b	B	U3
ionq_aria	IonQMS	GPi, GPi2
ionq_aria_full	IonQMSMax	GPi, GPi2
ionq_forte	IonQZZ	GPi, GPi2
quantinuum_zzmax	ZZMax	PhasedX, Rz
quantinuum_zzphase	ZZPhase	PhasedX, Rz
tket_tk2	TK2	PhasedX, Rz
rigetti_cz	CZ	Rz, Sx, X
aqt_rxx	Rxx	R, Rz
iqm_cz	CZ	R
clifford_t_cx	CX	H, S, T, Tdg
clifford_t_cz	CZ	H, S, T, Tdg
clifford_only	CX	H, S

The presets specify mathematical gate alphabets, not calibrated processors. A TK2 macro is one supplied primitive, and fixed discrete local alphabets are preserved even when they obstruct a continuous target family.

7 Reproduction

The source bundle includes the pinned environment manifests, current scripts, raw outputs, exact proof programs, and build inputs. The primary commands are:

```
python3 -m pytest tests/symbolic -q
python3 examples/decomposition/symbolic_referee_components.py \
 --output /tmp/components.json.gz
python3 examples/decomposition/symbolic_referee_peephole.py
python3 examples/decomposition/symbolic_referee_sdks.py \
 --output /tmp/sdks.json.gz
python3 examples/decomposition/symbolic_referee_rules.py \
```

```
--output /tmp/rules.json.gz
```

SDK comparisons use the separate Python-3.12 environment. Rule optimizers require the recorded Quartz build and Java-17 QUESO artifact; complete rule files, commands, and the harness patch are included. The builder regenerates coverage and comparison tables from the frozen records. Compilation of the paper needs `manuscript.tex` and `generated/`; the atlas sources retain their own relative `circuits/` and `programs/` paths.

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